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A cellular segmentation algorithm with fast customization

Common cellular segmentation models based on machine learning perform suboptimally for test images that differ greatly from training images. Cellpose 2.0 allows biologists to quickly train state-of-the-art segmentation models on their own imaging data. This was previously only possible using large, annotated datasets and required expert machine learning knowledge.

The problem

Many cellular-level analyses of biological systems, such as in situ RNA sequencing, developmental tracking and perturbation studies, require the segmentation of an image into single cells, inclusive of their cytoplasm and membrane[1]. However, cell segmentation is often technically difficult, particularly more so than the segmentation of other biological structures, such as nuclei[2]. Indeed, whereas nuclei are visually similar across specimens, cells can vary greatly in structural properties and shapes depending on the cell and tissue type, the imaging technique and the fluorescent markers used for their visualization. Further variation can be introduced by human error and differences in the ways that researchers segment the same images depending on the markers or cell features of interest. An ideal scenario would be for each user to create an optimized segmentation model for their own data, but there are several obstacles to this approach. State-of-the-art segmentation methods are based on deep learning models, and such models often require large training datasets to perform well; creating a large training dataset on a case-by-case basis would require substantial manual effort. Furthermore, machine learning experts may be needed to train each of these models.

The solution

We started with our original cell segmentation method, Cellpose[3]. This model has been trained on a dataset of annotated diverse cellular images and works well on data from various cell types and imaging modalities. Attempting to expand its use, we first tried to train Cellpose on an even larger set of data including several new large-scale annotated datasets (TissueNet[4] and LiveCell[5]). Unfortunately, the performance based on this larger dataset was poor, and we hypothesized the failure was due to differences in segmentation styles between different annotators. We concluded that specialized models are therefore necessary to perform best on a particular dataset.

Without further investigation, we did not know how many annotated cells would be necessary to train the model. Some of the datasets we worked with had over 200,000 manually drawn cells, which is a substantial effort requiring the employment of many annotators. However, we found a way to use many fewer annotations by pretraining the model on the original diverse Cellpose dataset. Starting with the pretrained Cellpose model[3], we found that only 1,000–2,000 manually drawn cells were required to achieve state-of-the-art performance for many cell types and imaging modalities. The number of manually drawn cells can be further reduced by training these models with a 'human-in-the-loop' approach, in which the models are incrementally improved as the user corrects the segmentations output by the model (Fig. 1). This reduces the manual effort to fewer than 200 cells, which corresponds to approximately 1 hour of user effort. We implemented this training protocol in a graphical user interface and tested it thoroughly ourselves to make it as effective and intuitive as possible, with the aim of providing a tool that can be easily and readily used by biologists. Preliminary feedback indicates that we achieved this goal, as many users have already been able to train their own models using our software.

The implications

Biologists often resort to segmenting cells manually because state-of-the-art segmentation methods are not built for their specific intended use. With Cellpose 2.0, users can now easily and rapidly create segmentation models for their own needs and then apply these models to all their datasets. This reduces manual effort and improves reproducibility because the models can be shared with the community and rerun by independent researchers.

One of the limitations of our method is common to all deep learning-based approaches: the model can only be as good at the task as the human training it. Generally, if a researcher cannot reliably and unambiguously segment cells, then the model will not be able to either. Another limitation relates to working with 3D volumes rather than the 2D data we used in this paper. Creating 3D segmentation models remains challenging because both the annotation and the models are more complicated in 3D. However, we expect increasing need for 3D models. The original Cellpose method had the ability to create 3D segmentations from models trained on 2D image data[3], and this approach should work better with user-trained 2D models in Cellpose 2.0. To do even better in 3D may also require novel 3D-specialized architectures and faster annotation strategies.

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EXPERT OPINION

"The authors clearly show that, with less annotation, they can achieve superior segmentation accuracy on two large datasets compared with two competing strategies. This is certainly valuable for this particular scenario as a reduced number of annotations is very attractive to the bioimage analysis community to expedite model training and cut down costs." **Alexandre Cunha, California Institute of Technology, Pasadena, CA, USA.**

FIGURE

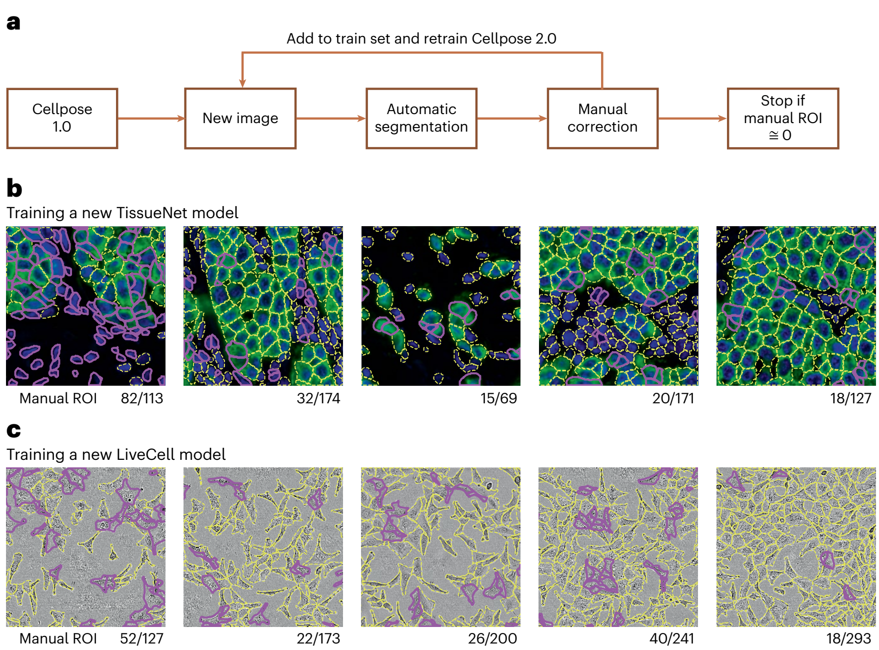

Fig. 1 | Human-in-the-loop training pipeline. a, Procedure for creating a personalized model in the Cellpose 2.0 graphical user interface. ROIs, regions of interest. **b**, A new TissueNet model was built by sequentially annotating five training images. After each image was segmented, the Cellpose model was retrained using all images annotated so far and initialized with the Cellpose parameters. On each new image, the latest model was applied and ROIs identified (yellow outlines), and the human curator added only the ROIs that were missed or incorrectly segmented by the automated method (purple outlines). **c**, Same as **b** for training a LiveCell model. Numbers below each image indicate the number of manually curated ROIs out of the total number of ROIs in the image. © 2022, Pachitariu, M. & Stringer, C., CCBY 4.0.

BEHIND THE PAPER

The initial Cellpose model and paper were published about two years ago. Cellpose attracted a large and diverse community of users, likely as a result of its versatility. While supporting this community, we started to see some cases in which the algorithm failed for certain types of data. Using our deep learning intuitions, we thought we could fix these failures by training Cellpose on an even bigger dataset. Coincidentally, some very big annotation efforts were being concluded at the time, and so we tried to incorporate their data and train one large model. This did not work so well, which was surprising. Nonetheless, the negative result motivated us to brainstorm alternative approaches. What if instead of going for bigger and bigger data, as was the norm in deep learning, we go for smaller and smaller data, but specific to each user application? How small can we go? This pushed our research into a completely new direction and eventually led to all the new developments in Cellpose 2.0. **M.P.**

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FROM THE EDITOR

"We were very excited to see the impact the original Cellpose tool and paper had on biological image analysis. When I read the Cellpose 2.0 paper, I was very pleased that it represented a giant leap from the original work. I am especially excited for the human-in-the-loop aspect of model training, which will undoubtedly expand the broad applicability of this already powerful approach." **Rita Strack, Senior Editor, *Nature Methods*.**